

Table 1: How generative AI helps key Python libraries in circuit design

Area	Python tools/ Libraries	How generative AI helps	Outcomes
Component selection	pandas, numpy, AI models	Suggests optimal components based on specs	BOM lists, alternatives, parametric choices
SPICE netlist generation	PySpice, skidl	Converts natural language to SPICE code	.cir netlists for simulation
Digital circuit design	nmigen, hdlparse	Creates combinational/sequential logic	Verilog/VHDL snippets, FSM diagrams
Circuit schematic generation	schemdraw, skidl	Auto-creates circuit diagrams from text prompts	Resistor networks, amplifiers, filters
Analog circuit design	PySpice, scipy.optimize	Generates topologies and optimises parameters	Op-amp designs, filters, oscillators
PCB layout suggestions	kicad_python, svgwrite	Provides placement/routing strategies	Suggested routing files, PCB scripts
Simulation automation	PySpice, matplotlib	Auto-runs multi-case simulations	Bode plots, time-domain results
Fault detection	sklearn, tensorflow	AI analyses signals to detect issues	Fault labels, warnings, insights
Optimisation of circuits	scipy, deap	Finds best values for components	Optimised RC, RLC, filter parameters
Power analysis	numpy, AI models	Predicts power issues, proposes fixes	Power curves, reduction suggestions
Thermal analysis	matplotlib, ML models	Estimates heat and suggests cooling	Heat maps, design tweaks
Signal processing support	scipy.signal, AI	Auto-designs filters and processing blocks	FIR/IIR designs, plots
EMI/EMC analysis	AI pattern models	Predicts noise sources, mitigation	Shielding suggestions, noise heatmaps
Embedded code generation	AI code models	Generates microcontroller code	Arduino, STM32 code
Testbench creation	pytest, AI models	Auto-generates tests for circuits	Test vectors, waveforms
Documentation creation	AI text generation	Creates design docs, reports	Circuit descriptions, manuals
Design validation	Custom Python validators	AI checks rule violations	DRC errors, corrections
Educational simulations	Python and AI chatbots	Generates learning simulations	Interactive circuit demos

Using generative AI with Python for design and simulation of circuits

Generative AI is no longer limited to chatbots and content writing. It can now be used for circuit design, as well as the modelling and simulation of high-performance engineering applications.

Python integrates a number of libraries that can be used to design and simulate circuits to help researchers as well as engineers predict their outcomes.

Generating circuits using SchemDraw

The SchemDraw (<https://schemdraw.readthedocs.io/>) script

can be generated with a prompt using ChatGPT or any other AI-based tool. This script is executed on Google Colab or any online Python-based notebook for generating the circuit.

```
!pip install PySpice schemdraw matplotlib

import schemdraw
import schemdraw.elements as el

with schemdraw.Drawing() as d:
    d += el.SourceV().up().label('5V')
    d += el.Resistor().right().label('1kΩ')
```